Solutions to MP352 Problem Set 3

1

P.1

There are several ways to check each of these matrices, but the way I'll do it here — which is probably the easiest way — is to see if they satisfy $\Lambda^T \eta \Lambda = \eta$, and $\det \Lambda = 1$. If Λ satisfies these, it's a Lorentz transformation; if it doesn't, it's not.

(1) $\Lambda_1^T \eta$ is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/\sqrt{3} & -\sqrt{2/3} \\ 0 & 0 & \sqrt{2/3} & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/\sqrt{3} & -\sqrt{2/3} \\ 0 & 0 & \sqrt{2/3} & 1/\sqrt{3} \end{pmatrix}$$

so

$$\Lambda_1^T \eta \Lambda_1 = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/\sqrt{3} & -\sqrt{2/3} \\ 0 & 0 & \sqrt{2/3} & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/\sqrt{3} & \sqrt{2/3} \\ 0 & 0 & -\sqrt{2/3} & 1/\sqrt{3} \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \left(\frac{1}{\sqrt{3}}\right)^2 + \left(-\sqrt{\frac{2}{3}}\right)^2 & \frac{1}{\sqrt{3}}\sqrt{\frac{2}{3}} - \sqrt{\frac{2}{3}}\frac{1}{\sqrt{3}} \\ 0 & 0 & \sqrt{\frac{2}{3}} \cdot \sqrt{\frac{1}{3}} - \sqrt{\frac{2}{3}}\frac{1}{\sqrt{3}} & \left(\sqrt{\frac{2}{3}}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Since we have only 0s in the first two rows & cols except for 1s on the diagonal, $\det \Lambda_1$ is just $\begin{vmatrix} 1/\sqrt{3} & -\sqrt{2/3} \\ \sqrt{2/3} & 1/\sqrt{3} \end{vmatrix} = \frac{1}{3} + \frac{2}{3} = 1$, so this is a Lorentz transformation (a rotation around the x-axis by an angle $= \cos^{-1}(1/\sqrt{3})$, in fact.)

(2) $\Lambda_2^T \eta$ is

$$\begin{pmatrix} \frac{3}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{2}{\sqrt{5}} & 0 & \frac{3}{\sqrt{5}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{3}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{2}{\sqrt{5}} & 0 & \frac{3}{\sqrt{5}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

so

$$\Lambda_2^T \eta \Lambda_2 = \begin{pmatrix} -\frac{3}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{2}{\sqrt{5}} & 0 & \frac{3}{\sqrt{5}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{3}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{2}{\sqrt{5}} & 0 & \frac{3}{\sqrt{5}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{9}{5} + \frac{4}{5} & 0 & -\frac{6}{5} + \frac{6}{5} & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{6}{5} + \frac{6}{5} & 0 & -\frac{4}{5} + \frac{9}{5} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The second and fourth rows & columns are all zeros except for 1s on the diagonal, so $\det \Lambda_2 = \begin{vmatrix} 3/\sqrt{5} & 2/\sqrt{5} \\ 2/\sqrt{5} & 3/\sqrt{5} \end{vmatrix} = \frac{9}{5} - \frac{4}{5} = 1$, so this is also a Lorentz transformation (a boost of $-2/3\,c$ in the y-direction).

(3) $\Lambda_3^T \eta$ is

$$\begin{pmatrix} \sqrt2 & -1/\sqrt2 & 0 & -1/\sqrt2 \\ -1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1/\sqrt2 & 0 & 1/\sqrt2 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -\sqrt2 & -1/\sqrt2 & 0 & -1/\sqrt2 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1/\sqrt2 & 0 & 1/\sqrt2 \end{pmatrix}$$

so

$$\Lambda_3^T \eta \Lambda_3 = \begin{pmatrix} -\sqrt2 & -1/\sqrt2 & 0 & -1/\sqrt2 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1/\sqrt2 & 0 & 1/\sqrt2 \end{pmatrix}\begin{pmatrix} \sqrt2 & -1 & 0 & 0 \\ -1/\sqrt2 & 1 & 0 & -1/\sqrt2 \\ 0 & 0 & 1 & 0 \\ -1/\sqrt2 & 1 & 0 & 1/\sqrt2 \end{pmatrix} = \begin{pmatrix} -2+\frac12+\frac12 & \sqrt2-\frac{1}{\sqrt2}-\frac{1}{\sqrt2} & 0 & \frac12-\frac12 \\ \sqrt2-\frac{1}{\sqrt2}-\frac{1}{\sqrt2} & -1+1+1 & 0 & -\frac{1}{\sqrt2}+\frac{1}{\sqrt2} \\ 0 & 0 & 1 & 0 \\ \frac12-\frac12 & -\frac{1}{\sqrt2}+\frac{1}{\sqrt2} & 0 & \frac12+\frac12 \end{pmatrix}$$

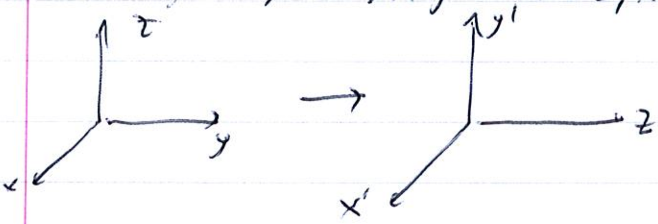
$$= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The determinant of this is a bit harder to compute, but it's doable:

$$\det \Lambda_3 = \begin{vmatrix} \sqrt2 & -1 & 0 & 0 \\ -\frac{1}{\sqrt2} & 1 & 0 & -1/\sqrt2 \\ 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt2} & 1 & 0 & \frac{1}{\sqrt2} \end{vmatrix} = \begin{vmatrix} \sqrt2 & -1 & 0 \\ -1/\sqrt2 & 1 & -1/\sqrt2 \\ -\frac{1}{\sqrt2} & 1 & 1/\sqrt2 \end{vmatrix} = \sqrt2 \begin{vmatrix} 1 & -\frac{1}{\sqrt2} \\ 1 & 1/\sqrt2 \end{vmatrix} + \begin{vmatrix} -\frac{1}{\sqrt2} & -\frac{1}{\sqrt2} \\ -\frac{1}{\sqrt2} & \frac{1}{\sqrt2} \end{vmatrix} = \sqrt2\left(-\frac{2}{\sqrt2}\right) + \left(-\frac{2}{\sqrt2}\right) = 1$$

so $\boxed{\text{it is indeed a Lorentz transformation}}$ as well.

(4) Checking $\Lambda_4^T \eta \Lambda_4 = \eta$ for this one is so easy I won't even write it down, but it's the determinant that gives us the answer: like Λ_1, the fact that the first two rows & columns are $\begin{pmatrix} 1&0&0&0 \\ 0&1&0&0 \\ 0&0&\cdot&\cdot \\ 0&0&\cdot&\cdot \end{pmatrix}$ tells us $\det \Lambda_4 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$. Since this is not 1, $\boxed{\Lambda_4 \text{ is not a Lorentz transformation}}$. You can see this by noticing that Λ_4 gives $t' = t$, $x' = x$, $y' = z$ and $z' = y$, which changes the orientation of our axes, i.e. takes the original right-handed (xyz) axes into a left-handed set $(x'z'y')$.



right-handed   left-handed

Since we do not allow Lorentz transformations to change the handedness of our coordinate system, this tells us why $\det \Lambda_4 = -1$ and why it's not a Lorentz transformation.

P.2

(a) $\Lambda^T \eta \Lambda = \eta$ is the starting point, so now let's multiply this from the right by Λ^{-1}: $(\Lambda^T \eta \Lambda)\Lambda^{-1} = \eta \cdot \Lambda^{-1} \Rightarrow \Lambda^T \eta = \eta \cdot \Lambda^{-1}$. Now multiply from the left by η^{-1}: $\eta^{-1}(\Lambda^T \eta) = \eta^{-1}(\eta \Lambda^{-1}) = \Lambda^{-1}$, so $\boxed{\Lambda^{-1} = \eta^{-1} \Lambda^T \eta,}$ as desired.

(b) This makes computing the inverse of any Lorentz transformation nearly trivial, as it's just matrix multiplication. We showed Λ_4 was not a Lorentz transformation, so we don't need to do that one. The first two are pretty easy:

$$\Lambda_1^{-1} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{3}} & \sqrt{\frac{2}{3}} \\ 0 & 0 & -\sqrt{\frac{2}{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}^T \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \boxed{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/\sqrt{3} & -\sqrt{2/3} \\ 0 & 0 & -\sqrt{2/3} & 1/\sqrt{3} \end{pmatrix}}$$

$$\Lambda_2^{-1} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3/\sqrt{5} & 0 & 2/\sqrt{5} & 0 \\ 0 & 1 & 0 & 0 \\ 2/\sqrt{5} & 0 & 3/\sqrt{5} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^T \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \boxed{\begin{pmatrix} \frac{3}{\sqrt{5}} & 0 & -\frac{2}{\sqrt{5}} & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{2}{\sqrt{5}} & 0 & \frac{3}{\sqrt{5}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}$$

Λ_3^{-1} takes a bit more work, so let's do it in detail:

$$\Lambda_3^{-1} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & -1 & 0 & 0 \\ -1/\sqrt{2} & 1 & 0 & -1/\sqrt{2} \\ 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 1 & 0 & 1/\sqrt{2} \end{pmatrix}^T \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -\sqrt{2} & -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} = \boxed{\begin{pmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}}.$$

If you like, you can check these explicitly by showing $\Lambda_1^{-1}\Lambda_1 = \Lambda_2^{-1}\Lambda_2 = \Lambda_3^{-1}\Lambda_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

P.3

(a) To check this one, let's let $e = 1$, $a = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $b = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$ and construct the multiplication table explicitly. Obviously, $ea = ae = a$ and $eb = be = b$; we need to see what a^2, b^2, ab and ba are:

$$a^2 = \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} - \frac{3}{4} + 2\left(-\frac{1}{2}\right)\left(\frac{i\sqrt{3}}{2}\right) = -\frac{1}{2} - \frac{i\sqrt{3}}{2} = b$$

$$b^2 = \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} - \frac{3}{4} + 2\left(-\frac{1}{2}\right)\left(-\frac{i\sqrt{3}}{2}\right) = -\frac{1}{2} + \frac{i\sqrt{3}}{2} = a$$

$ab = \left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)\left(-\frac{1}{2}-i\frac{\sqrt{3}}{2}\right) = \left(-\frac{1}{2}\right)^2 - \left(i\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4}+\frac{3}{4} = 1 = e$
and $ba = ab = e$ since a and b are just numbers. Thus, the multiplication table looks like:

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

So: (i) is it closed? Yes: all products of $\{e, a, b\}$ are also in $\{e, a, b\}$.

(ii) is it associative? Yes, because regular multiplication of complex numbers is associative.

(iii) Is there an identity? Yes, $e = 1$ is the identity.

(iv) Does every element have an inverse in the set? Yes: $e^{-1} = e$, $a^{-1} = b$ and $b^{-1} = a$.

Thus, all four properties are satisfied, so $\boxed{\text{this is indeed a group.}}$

(b) We can immediately see a problem with this one: the dot product between two three-dimensional real vectors is not a vector, it's a number. Thus, dotting two vectors in $\mathbb{R}^3$ does not yield a vector in $\mathbb{R}^3$, and so closure isn't satisfied. Therefore, it is $\boxed{\text{not a group}}$.

(c) This one is closed: if $\vec{A}$ and $\vec{B}$ are in $\mathbb{R}^3$, then $\vec{A}\times\vec{B}$ is as well, so at least closure is satisfied. But associativity is not: $\vec{A}\times(\vec{B}\times\vec{C})$ is not, in general, $(\vec{A}\times\vec{B})\times\vec{C}$. To show this, pick $\vec{A} = \vec{B} = \hat{e}_x$ and $\vec{C} = \hat{e}_y$. $\vec{B}\times\vec{C} = \hat{e}_x\times\hat{e}_y = \hat{e}_z$, so $\vec{A}\times(\vec{B}\times\vec{C}) = \hat{e}_x\times\hat{e}_z = -\hat{e}_y$. Here, $\vec{A}\times\vec{B} = \vec{0}$, so $(\vec{A}\times\vec{B})\times\vec{C} = \vec{0}$, and this gives an example of $(\vec{A}\times\vec{B})\times\vec{C} \neq \vec{A}\times(\vec{B}\times\vec{C})$. Thus, the cross-product is not an associative operation, and thus this is $\boxed{\text{not a group.}}$

P. 4

$GL(2,\mathbb{Z})$ consists of matrices of the form $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a, b, c and d are all integers and $ad - bc \neq 0$. Certainly closure is satisfied: $\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} = \begin{pmatrix} a_1a_2+b_1c_2 & a_1b_2+b_1d_2 \\ c_1a_2+d_1c_2 & c_1b_2+d_1d_2 \end{pmatrix}$

So if $a_1, b_1, c_1, d_1, a_2, b_2, c_2$ and d_2 are all integers, then so are $a_1a_2+b_1c_2$, $a_1b_2+b_1d_2$, $c_1a_2+d_1c_2$ and $c_1b_2+d_1d_2$. Matrix multiplication is associative already, so $(AB)C = A(BC)$ for any three matrices $A, B, C \in GL(2,\mathbb{Z})$. And $I = \begin{pmatrix}1&0\\0&1\end{pmatrix}$ is the identity matrix, also in $GL(2,\mathbb{Z})$ since 0 and 1 are integers.

However, the inverse of $\begin{pmatrix}a&b\\c&d\end{pmatrix}$ is

$$\begin{pmatrix}a&b\\c&d\end{pmatrix}^{-1} = \begin{pmatrix}\frac{d}{ad-bc} & \frac{-b}{ad-bc}\\ \frac{-c}{ad-bc} & \frac{a}{ad-bc}\end{pmatrix}$$

which might not be in $GL(2,\mathbb{Z})$, since it's possible that at least one of the entries is a non-integer. For example, consider $\begin{pmatrix}-1&1\\1&1\end{pmatrix}$; it's in $GL(2,\mathbb{Z})$, but its inverse is

$$\begin{pmatrix}-1&1\\1&1\end{pmatrix} = \begin{pmatrix}-1/2&1/2\\1/2&1/2\end{pmatrix}$$

which is not in $GL(2,\mathbb{Z})$. Since there are elements of this set whose inverses are not in the set, $\boxed{\text{it's not a group}}$.

However, all matrices in $SL(2,\mathbb{Z})$ have determinant 1, i.e. $ad-bc=1$. ~~Since the determinant of~~ AB is $(\det A)(\det B)$, then two matrices of unit determinant have a product that has determinant 1: thus, $A, B \in SL(2,\mathbb{Z}) \Rightarrow AB \in SL(2,\mathbb{Z})$. Again, associativity is automatic, and $\det\begin{pmatrix}1&0\\0&1\end{pmatrix} = 1$, so the identity matrix is in $SL(2,\mathbb{Z})$.

Now for the kicker: $ad-bc=1$ means that

$$\begin{pmatrix}a&b\\c&d\end{pmatrix}^{-1} = \begin{pmatrix}d&-b\\-c&a\end{pmatrix}$$

so if a, b, c and $d \in \mathbb{Z}$, then $d, -b, -c$ and a are in $\mathbb{Z}$. Therefore, all $SL(2,\mathbb{Z})$ matrices have inverses that are also in $SL(2,\mathbb{Z})$! This is the last property necessary, and so we see $\boxed{SL(2,\mathbb{Z}) \text{ is a group.}}$